

ELKANA JUANRO MANULLANG

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Bogor, Indonesia

Full-Stack Engineer with 2.5 years of hands-on experience in building scalable, data-driven solutions, specializing in PHP (Laravel), React/Next.js, and Supabase ecosystems. Proven track record of transforming manual operations into highly efficient Progressive Web Apps (PWA) and digital ecosystems, driving operational efficiency by up to 90% through custom-architected workflows. Adept at engineering ACID-compliant databases with Row Level Security (RLS), streamlining large-scale data pipelines to securely process more than 40,000 daily enterprise records, and integrating AI/LLM decision-support systems to accelerate business growth.

CORE THECNICAL SKILLS

Frontend	: React.js, Next.js, TypeScript, JavaScript, Tailwind CSS, Zustand, Redux, TanStack Query HTML5, CSS3
Backend and APIs	: Laravel, Node.js, Python, NestJS, RESTful APIs, Supabase, Supabase Realtime
Databases	: PostgreSQL, MySQL, Redis, Prisma ORM
Testing	: Postman (API), Cypress (E2E)
Systems and Data	: ETL Pipelines, Data Visualization, Business Process Automation
Practices	: PWA, Git, RESTful API, Microservices Architecture, Agile/Scrum, VPS Hosting

WORK EXPERIENCE

Celestara Eterna - Bandar Lampung, Lampung <i>Full Stack Engineer (Freelance)</i>	Mar 2024 – April 2026
- Boosted client operational efficiency by 90% and increased revenue by 27% by digitalizing manual workflows into custom e-commerce and a real-time Progressive Web App POS system utilizing Next.js 14, React.js, and PHP (Laravel). - Accelerated quantitative financial analysis efficiency by 80% by integrating Artificial Intelligence (AI) and Natural Language Processing (NLP) pipelines into the AltaQuant decision support system. - Architected scalable backend infrastructures and ACID-compliant PostgreSQL databases with Row Level Security (RLS), supporting high-traffic internal tools, seamless third-party APIs, and real-time inventory synchronization.	
PT.Telkom Indonesia - Bogor, Jawa Barat <i>Software Developer Internship</i>	June 2025 – Aug 2025
- Automated the processing of over 40,000 daily operational data records with perfect accuracy by engineering centralized web-based reporting Extract Transform Load (ETL) pipelines utilizing Python (Pandas) and PostgreSQL. - Reduced manual reporting time by over 75 percent by building real-time monitoring dashboards and automated daily calculation features utilizing Python (Flask) and Tailwind CSS. - Accelerated operational decision-making by 50 percent by delivering highly stable RESTful APIs, rigorously tested and validated using Postman, to feed intuitive frontend data visualizations.	
PT.Winnicode Garuda Teknologi - Bogor, Jawa Barat <i>Laravel Developer Internship</i>	Mar 2025 – Aug 2025
- Reduced content publishing delays by 40 percent by programming a real-time synchronization backend and RESTful APIs, systematically verified via Postman, utilizing Laravel, Reactjs (Typescript) within a microservices-style integration. - Improved server response time by 30 percent and prevented data inconsistencies by optimizing relational database queries and establishing secure authentication via Laravel Sanctum within a microservices-style integration.	
Christian Student Fellowship Institut Teknologi Sumatera - Lampung, Indonesia <i>Frontend Web Developer</i>	Jan 2024 - Jan 2025
- Improved website usability and accessibility across multiple devices by developing responsive user interfaces utilizing React.js, JavaScript, and Tailwind CSS. - Enhanced internal user experience by integrating robust RESTful APIs to centralize content access and streamline organizational communication channels.	

PROJECTS

Glorya Motor POS System Full Stack Engineer	April 2026 - Present
- Architected a real-time Progressive Web App (PWA) POS system utilizing Next.js 14 and Supabase, streamlining complex multi-role operational workflows for cashiers, mechanics, and owners. - Engineered ACID-compliant PostgreSQL database schemas and enforced Row Level Security (RLS) to manage real-time inventory locks, effectively preventing data inconsistencies and overselling. - Developed a secure Cashier Session management module featuring automated financial discrepancy calculations and real-time WebSocket notifications utilizing Supabase Realtime.	

Enterprise Wholesale POS System (Sistem POS Kasir Grosir Sembako) Full Stack Engineer	Jan 2026 – April 2026
- Reduced profit reporting errors to 0% by architecting a custom weighted-average HPP engine and FIFO batch stock system utilizing NestJS, Next.js 14, and Prisma ORM. - Eliminated manual membership reclassifications and accelerated daily reporting by 75% through automated background jobs utilizing BullMQ and PostgreSQL. - Implemented secure Role-Based Access Control (RBAC) with Redis-cached guards, enabling scalable multi-branch operations without necessitating code refactoring.	
LeafGenn - Automated Retail Leaflet System Full Stack Engineer	Oct 2025 – Jan 2026
- Boosted operational efficiency by over 90 percent and reduced production time from 12 hours to under 1 hour by engineering an automated end-to-end full-stack system utilizing React (TypeScript) and PHP (Laravel). - Eliminated layout formatting errors and ensured 100 percent data accuracy by developing dynamic visual generation algorithms and comprehensive automated data validation pipelines tested via Postman.	
SIMONI - Employee Performance Monitoring System Frontend Mobile Developer	Sept 2025 - Nov 2025
- Transformed manual WhatsApp reporting workflows into a centralized digital application, achieving full accountability for employee attendance and daily task tracking. - Improved organizational transparency across departments by implementing real-time geo-tagging features and intuitive user interfaces utilizing modern mobile development frameworks.	
AltaQuant - AI Trading Decision Support System Backend Engineer	Sept 2025 - Dec 2025
- Accelerated financial analysis efficiency by 80 percent by developing a hybrid backend architecture combining rule-based logic with Artificial Intelligence utilizing Python and a frontend with React Typescript. - Improved trading decision reliability and mitigated human bias by engineering a decision support system utilizing Natural Language Processing (NLP) pipelines and Large Language Model (LLM) integrations.	
STB Dismantling & Replacement Reporting System Software Developer	June 2025 - Aug 2025
- Automated the processing of over 40,000 daily operational data records with complete data integrity by engineering a centralized Extract Transform Load (ETL) pipeline utilizing Python Pandas and PostgreSQL. - Reduced manual reporting time by 75 percent by developing a centralized web-based backend and real-time monitoring dashboards utilizing Python Flask and Tailwind CSS tested via Postman.	
Next.js Developer Portfolio Frontend Developer	May 2025 - May 2025
- Improved digital discoverability and foundational SEO performance by developing a responsive Server-Side Rendered application utilizing Next.js. - Demonstrated production-ready deployment capabilities by integrating secure third-party APIs and seamlessly hosting the platform on Vercel for continuous rapid delivery.	
The Celestern Times - Modern News Platform Laravel Deeloper	Mar 2025 - Aug 2025
- Enabled scalable content management and reduced publishing delays by 40 percent by building a highly flexible user interface utilizing React.js paired with a PHP (Laravel) backend. - Streamlined editorial workflows by implementing secure Google OAuth authentication and role-based access control, guaranteeing secure data exchanges through Postman API testing.	

EDUCATION

INSTITUT TEKNOLOGI SUMATERA – Lampung, Indonesia	Aug 2022 – March 2026
<i>Bachelor’s Degree, Informatics Engineering</i>	

CERTIFICATES

- Software Engineer – HackerRank (2026)
- Frontend Developer (React) – HackerRank (2026)
- Intermediate Full-Stack JavaScript Web Developer – ITBOX (2024)

ORGANIZATIONAL EXPERIENCES

Informactic Engineering Student Association - Lampung Selatan, Indonesia	May 2024 - Jan 2025
<i>Head of Operational</i>	
- Led the operational planning and execution for a large-scale leadership program serving over 100 participants, achieving 100% operational fulfillment. - Enhanced cross-functional team collaboration by 30% by supervising 5+ sub-teams and directing workflows via digital collaboration tools.	